



**BAULKHAM HILLS  
HIGH  
SCHOOL**

**2020**

**YEAR 12  
HALF YEARLY  
ASSESSMENTS**

# Mathematics Extension 1

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## **General Instructions**

- Reading time – 5 minutes
- Working time – 1 hour and 30 minutes
- Write using black pen
- Write your NESA# and Teacher's name on your answer booklet
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Marks may be deducted for careless or badly arranged work

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## **Total marks: 53**

### **Section I – 5 marks** (pages 2 – 3)

- Attempt Questions 1 – 5
- Allow about 8 minutes for this section

### **Section II – 48 marks** (pages 4 – 7)

- Attempt Questions 6 – 8
- Allow about 1 hour and 22 minutes for this section

## Section I

5 marks

Attempt Questions 1 – 5

Allow about 8 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 5

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1  $\int \frac{dx}{4 + 9x^2} = ?$

(A)  $\frac{1}{3} \tan^{-1}\left(\frac{3x}{2}\right) + c$

(B)  $\frac{1}{6} \tan^{-1}\left(\frac{3x}{2}\right) + c$

(C)  $\frac{1}{3} \tan^{-1}\left(\frac{2x}{3}\right) + c$

(D)  $\frac{1}{6} \tan^{-1}\left(\frac{2x}{3}\right) + c$

- 2 The region bounded by the curve  $y = x^4$ , the  $y$ -axis and the line  $y = 16$  is rotated about the  $y$ -axis.

Which of the following expressions is correct for the volume of the solid of revolution created by this rotation?

(A)  $\pi \int_0^2 x^8 dx$

(B)  $\pi \int_0^{16} x^8 dx$

(C)  $\pi \int_0^2 \sqrt{y} dy$

(D)  $\pi \int_0^{16} \sqrt{y} dy$

3 What is the maximum value of  $6\cos\theta + 4\sin\theta$  ?

(A) 10

(B)  $2\sqrt{13}$

(C) 6

(D)  $2\sqrt{5}$

4 Which of the following is the primitive of  $\sin^2 x$  ?

(A)  $x - \frac{1}{2} \sin 2x + c$

(B)  $x + \frac{1}{2} \sin 2x + c$

(C)  $\frac{x}{2} - \frac{1}{4} \sin 2x + c$

(D)  $\frac{x}{2} + \frac{1}{4} \sin 2x + c$

5 Using the substitution  $u = 2x + 1$  ,  $\int (2x - 1)\sqrt{2x + 1} \, dx$  can be expressed as

(A)  $\frac{1}{2} \int \left( u^{\frac{3}{2}} + u^{\frac{1}{2}} \right) du$

(B)  $\frac{1}{2} \int \left( u^{\frac{3}{2}} - 2u^{\frac{1}{2}} \right) du$

(C)  $2 \int \left( u^{\frac{3}{2}} + u^{\frac{1}{2}} \right) du$

(D)  $2 \int \left( u^{\frac{3}{2}} - 2u^{\frac{1}{2}} \right) du$

**END OF SECTION I**

## Section II

48 marks

Attempt Questions 6 – 8

Allow about 1 hour 22 minutes for this section

Answer each question on the appropriate answer sheet. Each answer sheet must show your NES A#. Extra paper is available.

In Questions 6 to 8, your responses should include relevant mathematical reasoning and/or calculations.

**Marks**

**Question 6 (15 marks)** Use a *separate* answer sheet

(a) Differentiate the following functions with respect to  $x$

(i)  $f(x) = [\sin^{-1}(3x)]^3$  2

(ii)  $f(x) = \ln(\cos^{-1}x)$  2

(iii)  $f(x) = (x^2 + 1)\tan^{-1}x$  2

(b) Evaluate

(i)  $\int_{-2}^2 \frac{dx}{x^2 + 4}$  2

(ii)  $\int_0^1 \frac{dx}{\sqrt{4 - 3x^2}}$  3

(c) Solve  $2\sin\theta - \cos\theta = 1$ , for  $0^\circ \leq \theta \leq 360^\circ$ , correct to the nearest 4 minute, using the substitution  $t = \tan \frac{\theta}{2}$

**Question 7 (20 marks)** Use a *separate* answer sheet

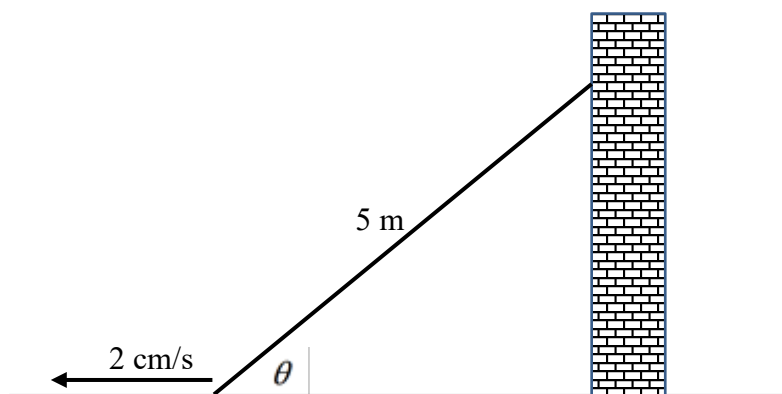
(a) Find the following integrals using the given substitution

(i)  $\int x(1 - x^2)^5 dx$  using the substitution  $u = 1 - x^2$  2

(ii)  $\int \frac{dx}{2x \ln x}$  using the substitution  $u = \ln x$  3

(iii)  $\int_0^{\frac{\pi}{4}} \frac{\sec^2 x}{\sqrt{1 + \tan x}} dx$  using the substitution  $u = 1 + \tan x$  3

(b) A 5 metre ladder is leaning against a wall on level ground. The base of the ladder is sliding away from the wall at a rate of 2 cm/s. 3



Find the rate at which the angle of inclination the ladder makes with the ground,  $\theta$ , is changing when the foot of the ladder is 3 metres from the base of the wall.

**Question 7 continues on page 6**

**Question 7 (continued)**

- (c) (i) Express  $\cos x - \sqrt{3}\sin x$  in the form  $R\cos(x + \alpha)$  where  $R > 0$  and  $0 < \alpha < \frac{\pi}{2}$  2

- (ii) Hence, or otherwise, solve; 3

$$\cos x - \sqrt{3}\sin x = 1 \quad \text{for} \quad 0 \leq x \leq 2\pi$$

- (d) (i) Show that  $\frac{d}{dx}(\tan^3 x) = 3(\sec^4 x - \sec^2 x)$  2

- (ii) Hence find  $\int \sec^4 x \, dx$  2

**End of Question 7**

**Question 8 (13 marks)** Use a *separate* answer sheet

- (a) Solve; 4

$$\sin 3x = \sin 2x \cos x \quad \text{for} \quad 0 \leq x \leq \frac{\pi}{2}$$

- (b) Find the volume of the solid generated when the curve  $y = \cos x + \sec x$  is rotated about the  $x$ -axis between  $x = 0$  and  $x = \frac{\pi}{3}$  4

- (c) (i) Show that; 2

$$\frac{d}{dx} \left( \tan^{-1} \frac{1-x}{1+x} \right) = -\frac{1}{1+x^2}$$

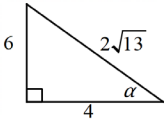
- (ii) Prove that for  $x > -1$ , 2

$$\tan^{-1} x + \tan^{-1} \frac{1-x}{1+x} = \frac{\pi}{4}$$

- (iii) Hence calculate  $\tan^{-1}(\sqrt{2} - 1)$  1

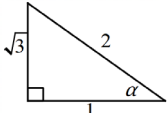
**End of paper**

**BAULKHAM HILLS HIGH SCHOOL**  
**YEAR 12 EXTENSION HALF YEARLY 2020 SOLUTIONS**

| Solution                                                                                                                                                                                                                                                      | Marks                                                                             | Comments |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------|
| <b>SECTION I</b>                                                                                                                                                                                                                                              |                                                                                   |          |
| <b>1. B -</b> $\int \frac{dx}{4 + 9x^2} = \frac{1}{3} \int \frac{3dx}{2^2 + (3x)^2}$ $= \frac{1}{3} \times \frac{1}{2} \tan^{-1} \left( \frac{3x}{2} \right) + c$ $= \frac{1}{6} \tan^{-1} \left( \frac{3x}{2} \right) + c$                                   | <b>1</b>                                                                          |          |
| <b>2. D -</b> $y = x^4 \Rightarrow x = y^{\frac{1}{4}}$ $V = \pi \int_0^{16} \left( y^{\frac{1}{4}} \right)^2 dy$ $= \pi \int_0^{16} y^{\frac{1}{2}} dy$ $= \pi \int_0^{16} \sqrt{y} dy$                                                                      | <b>1</b>                                                                          |          |
| <b>3. B -</b> $6\cos\theta + 4\sin\theta$<br>$= 2\sqrt{13} \sin(\theta + \alpha)$<br>$\therefore$ maximum value is $2\sqrt{13}$                                                                                                                               |  | <b>1</b> |
| <b>4. C -</b> $f'(x) = \sin^2 x$<br>$= \frac{1}{2}(1 - \cos 2x)$<br>$f(x) = \frac{1}{2} \left( x - \frac{1}{2} \sin 2x \right) + c$<br>$= \frac{x}{2} - \frac{1}{4} \sin 2x + c$                                                                              | <b>1</b>                                                                          |          |
| <b>5. B -</b> $\int (2x - 1) \sqrt{2x + 1} du$ $= \frac{1}{2} \int (u - 2) \sqrt{u} du$ $= \frac{1}{2} \int \left( u^{\frac{3}{2}} - 2u^{\frac{1}{2}} \right) du$ <div style="margin-left: 400px;"> <math>u = 2x + 1</math><br/> <math>du = 2dx</math> </div> | <b>1</b>                                                                          |          |

| SECTION II                                                                                                                                                                                                                                                                                                                                                                  |       |                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Solution                                                                                                                                                                                                                                                                                                                                                                    | Marks | Comments                                                                                                                                                                                                                                                                                   |
| QUESTION 6                                                                                                                                                                                                                                                                                                                                                                  |       |                                                                                                                                                                                                                                                                                            |
| <b>6(a) (i)</b> $f(x) = [\sin^{-1}(3x)]^3$<br>$f'(x) = 3[\sin^{-1}(3x)]^2 \times \frac{3}{\sqrt{1-9x^2}}$<br>$= \frac{9[\sin^{-1}(3x)]^2}{\sqrt{1-9x^2}}$                                                                                                                                                                                                                   | 2     | <b>2 marks</b><br>• Correct solution<br><b>1 mark</b><br>• Correctly differentiates $\sin^{-1}(3x)$<br>• Uses the chain rule with a valid attempt to differentiate $\sin^{-1}(3x)$                                                                                                         |
| <b>6(a) (ii)</b> $f(x) = \ln(\cos^{-1}x)$<br>$f'(x) = \frac{\left(-\frac{1}{\sqrt{1-x^2}}\right)}{\cos^{-1}x}$<br>$= -\frac{1}{\sqrt{1-x^2} \cos^{-1}x}$                                                                                                                                                                                                                    | 2     | <b>2 marks</b><br>• Correct solution<br><b>1 mark</b><br>• Correctly differentiates $\cos^{-1}x$<br>• Differentiates the log function with a valid attempt to differentiate $\cos^{-1}x$                                                                                                   |
| <b>6(a) (iii)</b> $f(x) = (x^2 + 1)\tan^{-1}x$<br>$f'(x) = (x^2 + 1)\left(\frac{1}{1+x^2}\right) + (\tan^{-1}x)(2x)$<br>$= 1 + 2x\tan^{-1}x$                                                                                                                                                                                                                                | 2     | <b>2 marks</b><br>• Correct solution<br><b>1 mark</b><br>• Correctly differentiates $\tan^{-1}x$<br>• Uses the product rule with a valid attempt to differentiate $\tan^{-1}x$                                                                                                             |
| <b>6(b) (i)</b> $\int_{-2}^2 \frac{dx}{x^2 + 4} = 2 \times \frac{1}{2} \left[ \tan^{-1}\left(\frac{x}{2}\right) \right]_0^2$<br>$= \tan^{-1} 1 - \tan^{-1} 0$<br>$= \frac{\pi}{4} - 0$<br>$= \frac{\pi}{4}$                                                                                                                                                                 | 2     | <b>2 marks</b><br>• Correct solution<br><b>1 mark</b><br>• Finds the correct primitive                                                                                                                                                                                                     |
| <b>6(b) (ii)</b> $\int_0^1 \frac{dx}{\sqrt{4-3x^2}} = \frac{1}{\sqrt{3}} \int_0^1 \frac{\sqrt{3} dx}{\sqrt{4-3x^2}}$<br>$= \frac{1}{\sqrt{3}} \left[ \sin^{-1} \frac{\sqrt{3}x}{2} \right]_0^1$<br>$= \frac{1}{\sqrt{3}} \left( \sin^{-1} \frac{\sqrt{3}}{2} - \sin^{-1} 0 \right)$<br>$= \frac{1}{\sqrt{3}} \left( \frac{\pi}{3} - 0 \right)$<br>$= \frac{\pi}{3\sqrt{3}}$ | 3     | <b>3 marks</b><br>• Correct solution<br><b>2 marks</b><br>• Finds the correct primitive<br>• Correctly evaluates definite integral using a primitive in the form of $k \sin^{-1} \frac{\sqrt{3}x}{2}$<br><b>1 mark</b><br>• Establishes the primitive as $k \sin^{-1} \frac{\sqrt{3}x}{2}$ |

| Solution                                                                                                                                                                                                                                                                                                   |  | Marks | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>6(c)</b> $2\sin\theta - \cos\theta = 1$ $2\left(\frac{2t}{1+t^2}\right) - \left(\frac{1-t^2}{1+t^2}\right) = 1$ $4t - 1 + t^2 = 1 + t^2$ $4t - 2 = 0$ $t = \frac{1}{2}$ $\tan\frac{\theta}{2} = \frac{1}{2}$ $\frac{\theta}{2} = 26^\circ 34'$ $\theta = 53^\circ 8'$ $\theta = 53^\circ 8', 180^\circ$ |  | 4     | <b>4 marks</b><br>• Correct solution<br><b>3 marks</b><br>• Correctly finds the solution $\theta = 53^\circ 8'$<br>• Transforms the trig equation into a polynomial equation and establishes $\theta = 180^\circ$<br><b>2 marks</b><br>• Transforms the trig equation into a polynomial equation<br><b>1 mark</b><br>• Substitutes the correct $t$ -results for $\sin\theta$ and $\cos\theta$<br>• establishes $\theta = 180^\circ$ as a solution |
| QUESTION 7                                                                                                                                                                                                                                                                                                 |  |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>7(a) (i)</b> $\int x(1-x)^5 dx = -\frac{1}{2} \int u^5 du$ $= -\frac{1}{12} u^6 + c$ $= -\frac{1}{12} (1-x^2)^6 + c$                                                                                                                                                                                    |  | 2     | <b>2 marks</b><br>• Correct solution<br><b>1 mark</b><br>• Rewrites integrand in terms of $u$                                                                                                                                                                                                                                                                                                                                                     |
| <b>7(a) (ii)</b> $\int \frac{dx}{2x \ln x} = \frac{1}{2} \int \frac{du}{u}$ $= \frac{1}{2} \ln u  + c$ $= \frac{1}{2} \ln \ln x  + c$                                                                                                                                                                      |  | 3     | <b>3 marks</b><br>• Correct solution<br><b>2 marks</b><br>• Finds the primitive in terms of $u$<br><b>1 mark</b><br>• Rewrites integrand in terms of $u$                                                                                                                                                                                                                                                                                          |
| <b>7(a) (iii)</b> $\int_0^{\frac{\pi}{4}} \frac{\sec^2 x}{\sqrt{1+\tan x}} dx = \int_1^2 \frac{2du}{\sqrt{u}}$ $= 2 \left[ \sqrt{u} \right]_1^2$ $= 2(\sqrt{2} - 1)$                                                                                                                                       |  | 3     | <b>3 marks</b><br>• Correct solution<br><b>2 marks</b><br>• Finds the primitive in terms of $u$<br>• Changes both the integrand and the limits in terms of $u$<br><b>1 mark</b><br>• Rewrites integrand in terms of $u$<br>• Expresses limits in terms of $u$                                                                                                                                                                                     |

| Solution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marks | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>7(b)</b> <math>\frac{dx}{dt} = \frac{1}{50} \text{ m/s}</math> <span style="float: right;"><math>\cos \theta = \frac{x}{5}</math></span></p> <p><math>\frac{d\theta}{dt} = \frac{dx}{dt} \times \frac{d\theta}{dx}</math> <span style="float: right;"><math>\theta = \cos^{-1} \frac{x}{5}</math></span></p> <p><math>= \frac{1}{50} \times \frac{-1}{\sqrt{25-x^2}}</math> <span style="float: right;"><math>\frac{d\theta}{dx} = \frac{-1}{\sqrt{25-x^2}}</math></span></p> <p><math>= \frac{-1}{50\sqrt{25-x^2}}</math></p> <p style="text-align: center;">when <math>x = 3</math>; <math>\frac{d\theta}{dt} = \frac{-1}{50\sqrt{25-9}}</math></p> <p style="text-align: center;"><math>= -\frac{1}{200} \text{ rad/s}</math></p> <p><math>\therefore</math> the angle of inclination is decreasing at a rate of <math>\frac{1}{200}</math> radians/second</p> | 3     | <p><b>3 marks</b></p> <ul style="list-style-type: none"> <li>• Correct solution, including correct units</li> </ul> <p><b>2 marks</b></p> <ul style="list-style-type: none"> <li>• Finds a correct expression for <math>\frac{d\theta}{dt}</math></li> </ul> <p><b>1 mark</b></p> <ul style="list-style-type: none"> <li>• Finds a correct expression for <math>\frac{d\theta}{dx}</math></li> <li>• Uses the chain rule to find an expression for <math>\frac{d\theta}{dt}</math></li> </ul> |
| <p><b>7 (c) (i)</b></p> <p><math>\cos x - \sqrt{3} \sin x</math></p> <p><math>= 2 \cos \left( x + \frac{\pi}{3} \right)</math></p> <div style="text-align: center;">  <p><math>\tan \alpha = \sqrt{3}</math></p> <p><math>\alpha = \frac{\pi}{3}</math></p> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2     | <p><b>2 marks</b></p> <ul style="list-style-type: none"> <li>• Correct solution</li> </ul> <p><b>1 mark</b></p> <ul style="list-style-type: none"> <li>• Finds <math>R</math> or <math>\alpha</math></li> </ul>                                                                                                                                                                                                                                                                               |
| <p><b>7(c) (ii)</b></p> <p><math>\cos x - \sqrt{3} \sin x = 1</math></p> <p><math>2 \cos \left( x + \frac{\pi}{3} \right) = 1</math></p> <p><math>\cos \left( x + \frac{\pi}{3} \right) = \frac{1}{2}</math></p> <p><math>x + \frac{\pi}{3} = \frac{\pi}{3}, \frac{5\pi}{3}</math></p> <p><math>x = 0, \frac{4\pi}{3}, 2\pi</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 3     | <p><b>3 marks</b></p> <ul style="list-style-type: none"> <li>• Finds all three solutions</li> </ul> <p><b>2 marks</b></p> <ul style="list-style-type: none"> <li>• Finds two values of <math>x</math></li> </ul> <p><b>1 mark</b></p> <ul style="list-style-type: none"> <li>• Finds two answers for <math>x + \frac{\pi}{3}</math></li> <li>• Finds one value of <math>x</math></li> </ul>                                                                                                   |
| <p><b>7(d) (i)</b></p> <p><math>\frac{d}{dx}(\tan^3 x) = 3(\tan^2 x)(\sec^2 x)</math></p> <p><math>= 3(\sec^2 x - 1)(\sec^2 x)</math></p> <p><math>= 3(\sec^4 x - \sec^2 x)</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2     | <p><b>2 marks</b></p> <ul style="list-style-type: none"> <li>• Correct solution</li> </ul> <p><b>1 mark</b></p> <ul style="list-style-type: none"> <li>• Uses chain rule</li> </ul>                                                                                                                                                                                                                                                                                                           |
| <p><b>7(d) (ii)</b></p> <p><math>3 \int (\sec^4 x - \sec^2 x) dx = \tan^3 x</math></p> <p><math>3 \int \sec^4 x dx = \tan^3 x + 3 \int \sec^2 x dx</math></p> <p style="text-align: center;"><math>= \tan^3 x + 3 \tan x + c</math></p> <p><math>\int \sec^3 x dx = \frac{1}{3} \tan^3 x + \tan x + c</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2     | <p><b>2 marks</b></p> <ul style="list-style-type: none"> <li>• Correct solution</li> </ul> <p><b>1 mark</b></p> <ul style="list-style-type: none"> <li>• Uses part (i) to write an expression for <math>\int \sec^4 x dx</math></li> </ul>                                                                                                                                                                                                                                                    |

| Solution   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| QUESTION 8 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 8(a)       | $\sin 3x = \sin 2x \cos x$ $= \frac{1}{2}(\sin 3x + \sin x)$ $2\sin 3x = \sin 3x + \sin x$ $\sin 3x = \sin x$ <p><b>Q1, Q2</b></p> $3x = x, \pi - x$ $2x = 0 \quad \text{or} \quad 4x = \pi$ $x = 0 \quad \quad x = \frac{\pi}{4}$                                                                                                                                                                                                                                                                                                                      | 4     | <p><b>4 marks</b></p> <ul style="list-style-type: none"> <li>• Correct solution</li> </ul> <p><b>3 marks</b></p> <ul style="list-style-type: none"> <li>• Correctly finds one solution for <math>x</math></li> </ul> <p><b>2 marks</b></p> <ul style="list-style-type: none"> <li>• establishes <math>\sin 3x = \sin x</math></li> <li>• states one correct solution for <math>x</math></li> </ul> <p><b>1 mark</b></p> <ul style="list-style-type: none"> <li>• Uses “products to sum” formula to simplify <math>\sin 3x + \sin x</math></li> </ul>                                                                                                                                                                                                                                                                                                              |
| 8(b)       | $V = \pi \int_0^{\frac{\pi}{3}} (\cos x + \sec x)^2 dx$ $= \pi \int_0^{\frac{\pi}{3}} (\cos^2 x + 2 + \sec^2 x) dx$ $= \pi \int_0^{\frac{\pi}{3}} \left\{ \frac{1}{2}(1 + \cos 2x) + 2 + \sec^2 x \right\} dx$ $= \pi \int_0^{\frac{\pi}{3}} \left( \frac{5}{2} + \frac{1}{2} \cos 2x + \sec^2 x \right) dx$ $= \pi \left[ \frac{5x}{2} + \frac{1}{4} \sin 2x + \tan x \right]_0^{\frac{\pi}{3}}$ $= \pi \left( \frac{5\pi}{6} + \frac{\sqrt{3}}{8} + \sqrt{3} - 0 \right)$ $= \pi \left( \frac{5\pi}{6} + \frac{9\sqrt{3}}{8} \right) \text{ units}^3$ | 4     | <p><b>4 marks</b></p> <ul style="list-style-type: none"> <li>• Correct solution</li> </ul> <p><b>3 marks</b></p> <ul style="list-style-type: none"> <li>• Finds the primitive function</li> <li>• Successfully evaluates primitive that has been obtained by correct integration of a trig function found via a relevant process</li> </ul> <p><b>2 marks</b></p> <ul style="list-style-type: none"> <li>• Uses the identity <math>\cos^2 x = \frac{1}{2}(1 + \cos 2x)</math> to rewrite integrand into an easily handled form</li> <li>• Finds a primitive that has been obtained by correct integration of a trig function found via a relevant process</li> </ul> <p><b>1 mark</b></p> <ul style="list-style-type: none"> <li>• finds the primitive of <math>\sec^2 x</math></li> </ul> <p><i>Note: decimal approximation of 14.34624319 is acceptable</i></p> |
| 8(c) (i)   | $\frac{d}{dx} \left( \tan^{-1} \frac{1-x}{1+x} \right) = \frac{\frac{(1+x)(-1) - (1-x)(1)}{(1+x)^2}}{1 + \left( \frac{1-x}{1+x} \right)^2}$ $= \frac{-2}{(1+x)^2 + (1-x)^2}$ $= \frac{-2}{2 + 2x^2}$ $= -\frac{1}{1+x^2}$                                                                                                                                                                                                                                                                                                                               | 2     | <p><b>2 marks</b></p> <ul style="list-style-type: none"> <li>• Correct solution</li> </ul> <p><b>1 mark</b></p> <ul style="list-style-type: none"> <li>• Correctly uses the differentiation rule for <math>\tan^{-1} f(x)</math></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Solution   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Marks | Comments                                                                                                                                                                                                                                                                             |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| QUESTION 8 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |                                                                                                                                                                                                                                                                                      |
| 8(c) (ii)  | $\frac{d}{dx} \left( \tan^{-1} x + \tan^{-1} \frac{1-x}{1+x} \right) = \frac{1}{1+x^2} - \frac{1}{1+x^2}$ $= 0$ <p><math>\therefore</math> as both functions are continuous for <math>x &gt; -1</math>,</p> $\tan^{-1} x + \tan^{-1} \frac{1-x}{1+x} = c$ <p>let <math>x = 0</math>; <math>\tan^{-1} x + \tan^{-1} \frac{1-x}{1+x} = \tan^{-1} 0 + \tan^{-1} 1</math></p> $= \frac{\pi}{4}$ <p>thus <math>\tan^{-1} x + \tan^{-1} \frac{1-x}{1+x} = \frac{\pi}{4}</math>, for <math>x &gt; -1</math></p> <p><i>OR</i></p> $\tan \left( \tan^{-1} x + \tan^{-1} \frac{1-x}{1+x} \right) = \frac{x + \frac{1-x}{1+x}}{1 - x \left( \frac{1-x}{1+x} \right)}$ $= \frac{x(1+x) + (1-x)}{(1+x) - x(1-x)}$ $= \frac{x^2 + 1}{1+x^2}$ $= 1$ <p><math>\therefore \tan^{-1} x + \tan^{-1} \frac{1-x}{1+x} = \tan^{-1} 1</math></p> $= \frac{\pi}{4}$ | 2     | <b>2 marks</b> <ul style="list-style-type: none"> <li>Correct proves result</li> </ul> <b>1 mark</b> <ul style="list-style-type: none"> <li>Explains why the function is a constant</li> <li>Shows that</li> </ul> $\tan \left( \tan^{-1} x + \tan^{-1} \frac{1-x}{1+x} \right) = 1$ |
| 8(c) (iii) | <p>let <math>x = \sqrt{2} - 1</math>;</p> $\tan^{-1}(\sqrt{2} - 1) + \tan^{-1} \left( \frac{2 - \sqrt{2}}{\sqrt{2}} \right) = \frac{\pi}{4}$ $\tan^{-1}(\sqrt{2} - 1) + \tan^{-1}(\sqrt{2} - 1) = \frac{\pi}{4}$ $2 \tan^{-1}(\sqrt{2} - 1) = \frac{\pi}{4}$ $\tan^{-1}(\sqrt{2} - 1) = \frac{\pi}{8}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1     | <b>1 mark</b> <ul style="list-style-type: none"> <li>Correct exact value</li> </ul>                                                                                                                                                                                                  |